

Python Operators

Operators are special symbols used to perform operations on variables and values.

Example:

```
a = 10
b = 3

print(a + b)
```

Output:

13

1. Arithmetic Operators

Used to perform mathematical calculations.

Operator	Name	Example	Result
+	Addition	10 + 3	13
-	Subtraction	10 - 3	7
*	Multiplication	10 * 3	30
/	Division	10 / 3	3.333
%	Modulus	10 % 3	1
//	Floor Division	10 // 3	3
**	Power	10 ** 3	1000

Example

```
a = 10
b = 3

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
print("Modulus =", a % b)
print("Floor Division =", a // b)
print("Power =", a ** b)
```

2. Comparison Operators

Used to **compare two values**. The result is either `True` or `False`.

Operator	Meaning	Example
==	Equal to	10 == 10 → True

Operator	Meaning	Example
!=	Not equal to	10 != 5 → True
>	Greater than	10 > 5 → True
<	Less than	10 < 5 → False
>=	Greater than or equal to	10 >= 10 → True
<=	Less than or equal to	5 <= 10 → True

Example

```
a = 10
b = 5

print(a > b)
print(a == b)
print(a != b)
```

Output:

```
True
False
True
```

3. Assignment Operators

Used to **assign values** to variables.

Operator	Example	Equivalent
=	a = 10	a = 10
+=	a += 5	a = a + 5
-=	a -= 5	a = a - 5
*=	a *= 5	a = a * 5
/=	a /= 5	a = a / 5
%=	a %= 5	a = a % 5

Example

```
a = 10
a += 5

print(a)
```

Output:

```
15
```

4. Logical Operators

Used to combine **conditions**.

Operator	Meaning
and	Both conditions must be true
or	At least one condition must be true
not	Reverses the result

Example

```
age = 20

print(age >= 18 and age <= 60)
print(age < 18 or age > 60)
print(not (age >= 18))
```

Output:

```
True
False
False
```

5. Membership Operators

Used to check whether a value is present in a sequence such as a list or string.

Operator	Meaning
in	Value exists
not in	Value does not exist

Example

```
students = ["Arun", "Bala", "Karthik"]

print("Arun" in students)
print("Rahul" not in students)
```

Output:

```
True
True
```

6. Identity Operators

Used to check whether two variables refer to the **same object**.

Operator	Meaning
is	Same object

Operator	Meaning
is not	Different objects

Example

```
a = [1, 2, 3]
b = a

print(a is b)
```

Output:

True

7. Bitwise Operators

Used to perform operations on numbers at the **binary/bit level**.

Operator	Name
&	AND
	OR
^	XOR
~	NOT
<<	Left Shift
>>	Right Shift

Example

```
a = 5
b = 3

print(a & b)
print(a | b)
```

Output:

1
7

Quick Summary

Python mainly has these **7 types of operators**:

1. **Arithmetic** → + - * / % // **
2. **Comparison** → == != > < >= <=
3. **Assignment** → = += -= *= /= %=
4. **Logical** → and or not

5. **Membership** → `in`, `not in`
6. **Identity** → `is`, `is not`
7. **Bitwise** → `&` `|` `^` `~` `<<` `>>`

These are the main operators you should know for **basic Python programming and exams**.